

Week 1 · Laboratory Problems — build the open loop in Simulink

- Course: Sensor Signal Processing and Fusion · Department of Autonomous Vehicle System Engineering
- Time: **one hour**, immediately after the Week 1 lecture hour
- Three problems, in order. Each one starts where the previous one stopped.

What this hour is for, and why the models are not simply given

The lecture builds `w01_openloop.slx` **from a script**. Running one command produces a finished model, and reading that script is a good way to learn what the finished model contains. It is not a good way to learn how to build one.

This hour reverses that. The hull is provided, because integrating `otter.m` is not the subject of Week 1 and cannot be assembled from library blocks in an hour. Everything else — what the command is, how it reaches the plant, which of the twelve states matter, how a result is logged — is built by hand.

Every problem ends with a **number that was measured in the lecture**. A model that reproduces that number is doing the same physics as the lecture's model, whatever it looks like on the canvas. There is no single correct diagram.

Before starting

```
cd lectures/w01_simulink/problems
w01_P1_start           % creates w01_P1.slx – the hull, and nothing else
```

The solver is already set to fixed-step `ode4` at `h = 0.02` s. **Do not change it**. The numbers below were measured with those settings.

Check work at any time:

```
w01_check(1)           % or w01_check(1, 'w01_P1_yourname')
```

The checker runs the model, compares it with the lecture, and prints `PASS` or `FAIL` for each test together with the value it expected.

★ One requirement on every model

The model must contain **one To Workspace block named `xlog`**, with Save format **Structure With Time**, fed by the plant's twelve-state output.

That is the only thing the checker needs. Requiring the raw twelve states rather than a tidy six-column log is deliberate: picking u , v , r , x , y and ψ out of the vector is the part of Week 1 worth being able to do without looking it up.

Problem 1 · The open loop (25 minutes)

Build. A constant command into the hull, and a log coming out.

```
Constant [n0 ; n0] → Otter USV → To Workspace (xlog)
```

Predict before running. At steady state the thrust balances linear surge damping:

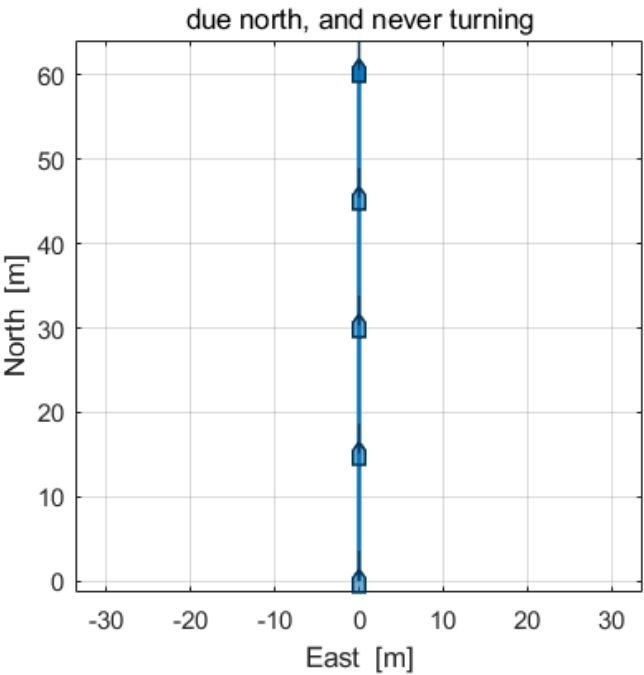
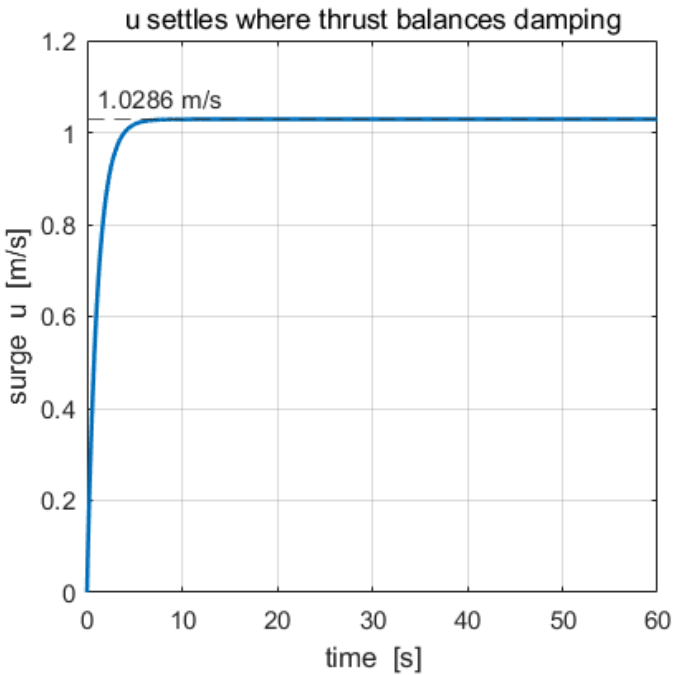
$$2\,k_{pos}\,n\,|n| = X_u\,u \quad \implies \quad u = \frac{2(0.01108)(60)(60)}{77.554}$$

Write the number down before pressing Run.

Verify. `w01_check(1)` — terminal u , and that v and r stay at zero.

What the checker expects	
terminal surge speed	1.0286 m/s
sway velocity v	exactly 0
yaw rate r	exactly 0

What a correct model produces



Reading the figure	
left	u rises with one time constant and settles on the dashed line at 1.0286 m/s. It does not overshoot: there is no loop, so nothing can
right	the track is a straight line due north, and every hull silhouette on it points the same way
the check	if the track bends at all, the two Constant entries are not equal

The point. Two propellers, equal shaft speeds. The yaw moment is $N = y_p (T_{left} - T_{right})$, so equal speeds produce no turn at all, and the run is a test of surge alone. The command is a **2-vector**: `otter.m` expects $n = [n_L ; n_R]$, and a scalar is accepted by the Constant block and rejected by the plant.

Problem 2 · The manoeuvre (20 minutes)

Build. Replace the constant by a schedule of time: straight, port, straight, starboard, straight.

```
Clock → MATLAB Function → Reshape (1-D array) → Otter USV → xlog
```

Decide the signs first. This is the whole problem, and it is decided before any block is placed:

$$N = y_p (T_{left} - T_{right}), \quad y_p = 0.395 \text{ m}$$

More thrust on the **left** turns the bow to **starboard**. A port turn therefore **slows the left propeller**:

phase	command
straight	$n = [n_0 ; n_0]$
to port	$n = [n_0 - \delta n ; n_0 + \delta n]$
to starboard	$n = [n_0 + \delta n ; n_0 - \delta n]$

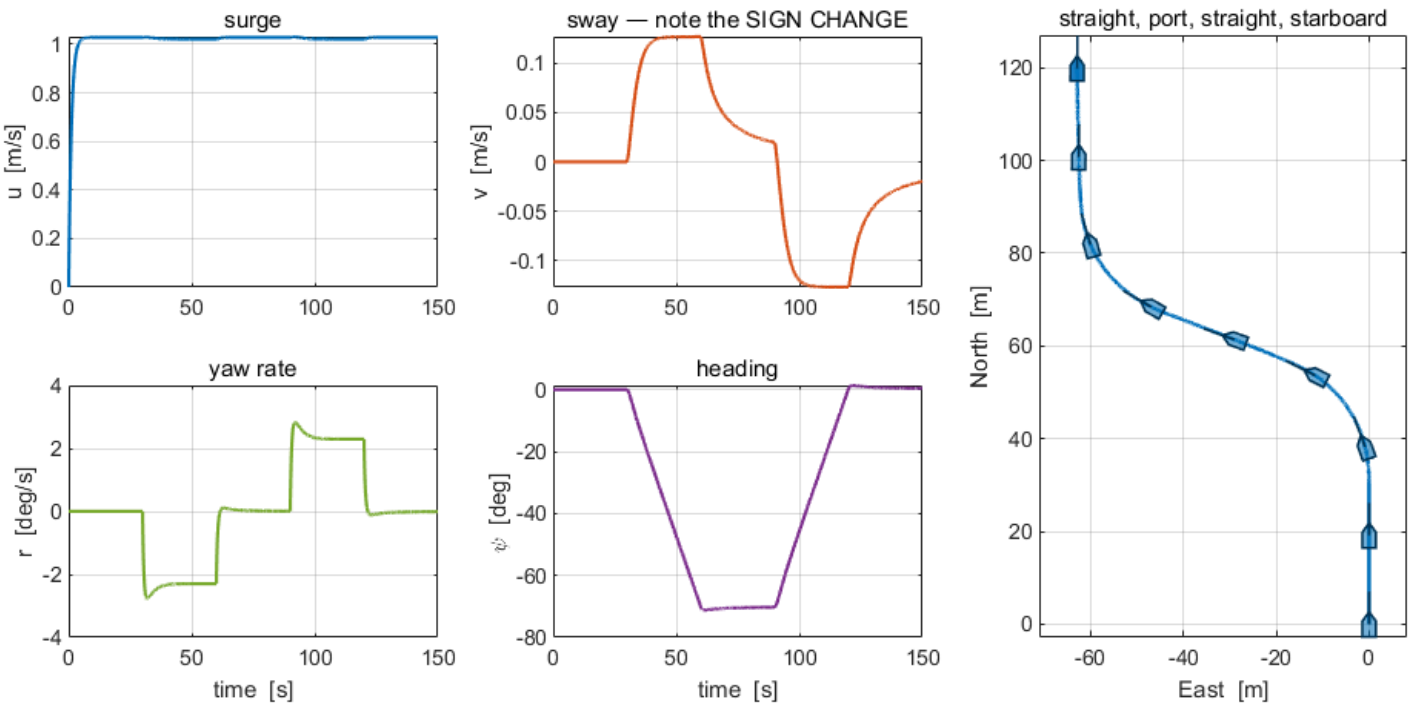
Nothing ever goes astern. At $n_0 = 60$ and $\delta n = 3.5$ the slower propeller still pushes ahead at **35.37** N while the faster one pushes at **44.68** N.

Verify. `w01_check(2)` .

What the checker expects	
yaw rate in the port turn	-2.2942 deg/s
heading change, port then starboard	-70.2 then +70.6 deg
sway v in the two turns	+0.1264 then -0.1264 m/s

Getting the sign backwards is visible immediately: the heading change comes out **+70** deg where the checker wants **-70** deg.

What a correct model produces



Reading the figure	
u	barely moves. The turns cost about 0.007 m/s of speed and nothing else
v	rises to +0.126 in the first turn and falls to −0.126 in the second. The sign change is the point of the problem
r	two flat plateaus of opposite sign, one per turn, with the transient at each phase boundary
ψ	a ramp down to −70 deg, a hold, then a ramp back. A ramp UP first means the sign is reversed
track	the S-shape, with the hull drawn along it so the heading is visible where the track alone would not show it

⚠ The Reshape block is not decoration

A MATLAB Function block whose output is written `n = [nL; nR]` produces a **2 × 1 matrix** signal. The plant does not care. The log does: one matrix input makes the whole To Workspace record three-dimensional, and the checker then finds one column where it expects twelve.

The point. $\mathbf{Y}$ is exactly zero at every instant — both propellers face forward, so no combination of them has a component across the hull. Yet v is non-zero in **both** turns and **changes sign** between them. That sway is produced by the hull rotating, through the Coriolis term, and not by any side force. It is the crab angle, and Week 3 has to steer around it.

Problem 3 · The vessel in a current (15 minutes)

Build. Nothing. The model of Problem 1 is used unchanged.

That is the exercise: work out **why** nothing needs to be added before running anything.

Read `otter.m` **lines 79–81 and line 204.**

$$u_c = V_c \cos(\beta_c - \psi), \quad v_c = V_c \sin(\beta_c - \psi), \quad \nu_r = \nu - [u_c \ v_c \ 0 \ 0 \ 0]^T$$

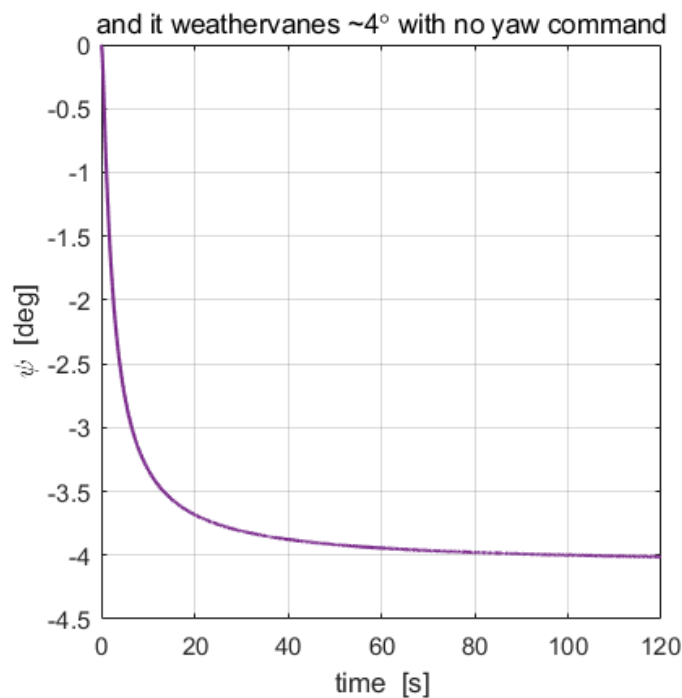
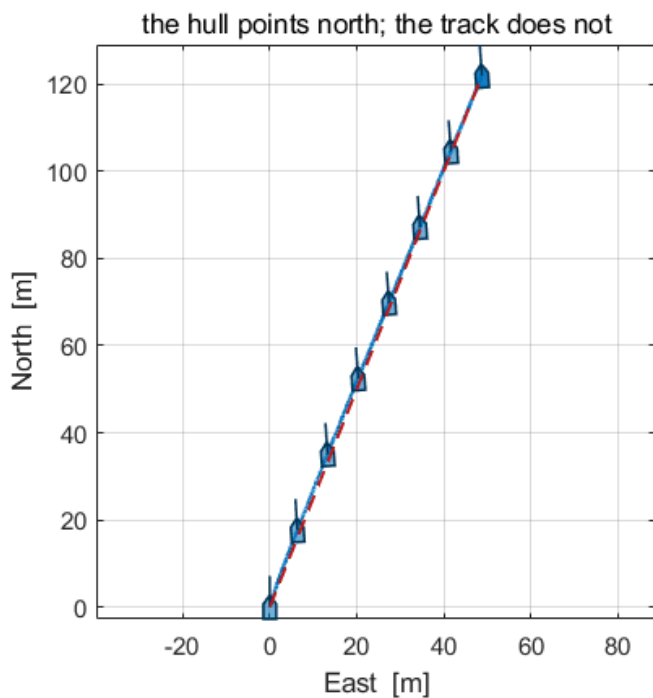
$$\dot{\eta} = \mathbf{J}(\eta) \nu \quad \text{— with } \nu, \text{ not } \nu_r$$

Predict before running. With $V_c = 0.5$ m/s on the beam and the vessel making **1.0286** m/s through the water, the velocity over the ground is the vector sum, so the drift is roughly $\arctan(0.5/1.0286) = 25.9^\circ$.

Verify. `w01_check(3)` — it sets $V_c = 0.5$ m/s and $\beta_c = 90^\circ$ itself.

What the checker expects	
ground speed	1.1091 m/s
track direction from North	21.804 deg
drift, track minus heading	25.809 deg

What a correct model produces



Reading the figure	
left	the hull silhouettes point north while the track leans east . The dashed red line is the straight track from start to finish, at 21.8°
right	ψ drifts about -4° over the run, with no yaw command anywhere in the model
the check	if the hull silhouettes lean over with the track, ψ is being drawn from the wrong state — the heading is index 12, not the direction of travel

The point. A current is a **velocity, not a force**. Every hydrodynamic term is computed from $\boldsymbol{\nu}_r$, the velocity through the water, so the forces on the hull do not change. The position integrates $\boldsymbol{\nu}$, so the vessel is carried along by the water. That single asymmetry is why a current moves a vessel without pushing it.

The predicted drift of **25.9°** and the measured **25.81°** agree, but the ground speed comes out below the naive vector sum of **1.144 m/s**: the hull weathervanes about **4°** into the flow with no yaw command anywhere in the model, which turns part of the current from beam-on to head-on.

Marking

	Weight	What is being marked
Problem 1	30	<code>w01_check(1)</code> passes, and the predicted 1.0286 m/s was written down before running
Problem 2	40	<code>w01_check(2)</code> passes; the sign argument for the port turn is stated in one sentence
Problem 3	30	<code>w01_check(3)</code> passes; the answer to "why was nothing added to the model" is stated

A model that fails a check but whose written reasoning is right earns more than one that passes with no reasoning. The numbers are there to make the reasoning testable, not to replace it.

If something goes wrong

Symptom	Cause	Fix
The model has no To Workspace block whose variable name is xlog	the block exists but its variable name is still <code>simout</code>	open it and set the variable name to <code>xlog</code>
xlog is not a Structure With Time	Save format left at <code>Array</code> or <code>Timeseries</code>	set Save format to <code>Structure With Time</code>
xlog has 1 column. Feed the To Workspace the FULL 12-state vector	a matrix signal reached the log	insert a Reshape set to <code>1-D array</code> after the MATLAB Function
Dimension error at the plant input	the command is a scalar	the Constant must be <code>[n0 ; n0]</code>
Heading changes the wrong way in Problem 2	the port turn speeds the left propeller up	port turn slows the left one — $N = y_p(T_{left} - T_{right})$
Everything runs but every number is slightly off	the solver was changed	fixed-step <code>ode4</code> , <code>h = 0.02</code> s

Reference solutions are in `../solutions/` . Read them **after** attempting the problem: each one explains not only what to build but why the alternatives were rejected.